

## SAFETY DATA SHEET

Creation Date 26-Sep-2009

Revision Date 16-Feb-2024

Revision Number 6

### 1. Identification

**Product Name** Dibutyl phosphate

**Cat No. :** AC406640000; AC406640025; AC406642500; AC406645000

**CAS No** 107-66-4  
**Synonyms** Dibutyl Acid Phosphate; Dibutyl Hydrogen Phosphate.

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

**Company**

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
**Emergency Number US:**001-201-796-7100 / **Europe:** +32 14 57 52 99  
**CHEMTREC** Tel. No. **US:**001-800-424-9300 / **Europe:**001-703-527-3887

### 2. Hazard(s) identification

**Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                   |              |
|-----------------------------------|--------------|
| Skin Corrosion/Irritation         | Category 1 B |
| Serious Eye Damage/Eye Irritation | Category 1   |
| Carcinogenicity                   | Category 2   |

**Label Elements****Signal Word**

Danger

**Hazard Statements**

Causes severe skin burns and eye damage  
Suspected of causing cancer

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Use personal protective equipment as required  
Do not breathe dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling

**Response**

Immediately call a POISON CENTER or doctor/physician

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

**Skin**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower  
Wash contaminated clothing before reuse

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

**Ingestion**

IF SWALLOWED: Rinse mouth. DO NOT induce vomiting

**Storage**

Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

### 3. Composition/Information on Ingredients

| Component         | CAS No   | Weight % |
|-------------------|----------|----------|
| Dibutyl phosphate | 107-66-4 | <=100    |

### 4. First-aid measures

|                                            |                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.                                                                                                                                                                                       |
| <b>Skin Contact</b>                        | Immediate medical attention is required. Wash off immediately with plenty of water for at least 15 minutes.                                                                                                                                                                                                             |
| <b>Inhalation</b>                          | Remove to fresh air. If breathing is difficult, give oxygen. Immediate medical attention is required. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                          |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. . Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated                       |

Notes to Physician Treat symptomatically

## 5. Fire-fighting measures

**Suitable Extinguishing Media** Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam.

**Unsuitable Extinguishing Media** No information available

**Flash Point** 180 °C / 356 °F

**Method -** No information available

**Autoignition Temperature** 420 °C / 788 °F

**Explosion Limits**

**Upper** No data available

**Lower** No data available

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

**Specific Hazards Arising from the Chemical**

Corrosive material. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Oxides of phosphorus.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
3

**Flammability**  
1

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions** Use personal protective equipment as required. Ensure adequate ventilation. Avoid contact with skin and eyes.

**Environmental Precautions** See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up** Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Do not let this chemical enter the environment.

## 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Use only under a chemical fume hood. Do not get in eyes, on skin, or on clothing. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

**Storage.** Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Corrosives area. Do not store in metal containers. Incompatible Materials. Strong oxidizing agents.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component         | ACGIH TLV                        | OSHA PEL                                                                                                                                                              | NIOSH                                                                                               | Mexico OEL (TWA)         |
|-------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------|
| Dibutyl phosphate | TWA: 5 mg/m <sup>3</sup><br>Skin | (Vacated) TWA: 1 ppm<br>(Vacated) TWA: 5 mg/m <sup>3</sup><br>(Vacated) STEL: 2 ppm<br>(Vacated) STEL: 10 mg/m <sup>3</sup><br>TWA: 1 ppm<br>TWA: 5 mg/m <sup>3</sup> | IDLH: 30 ppm<br>TWA: 1 ppm<br>TWA: 5 mg/m <sup>3</sup><br>STEL: 2 ppm<br>STEL: 10 mg/m <sup>3</sup> | TWA: 5 mg/m <sup>3</sup> |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.

**Personal Protective Equipment****Eye/face Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection**

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended Filter type:**

Organic gases and vapours filter. Type A. Brown. conforming to EN14387.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                                    |                            |
|----------------------------------------------------|----------------------------|
| Physical State                                     | Liquid                     |
| Appearance                                         | Colorless                  |
| Odor                                               | Characteristic             |
| Odor Threshold                                     | No information available   |
| pH                                                 | 1.4 (10%)                  |
| Melting Point/Range                                | -13 °C / 8.6 °F            |
| Boiling Point/Range                                | 250 °C / 482 °F @ 760 mmHg |
| Flash Point                                        | 180 °C / 356 °F            |
| Evaporation Rate                                   | No information available   |
| Flammability (solid,gas)                           | Not applicable             |
| Flammability or explosive limits                   |                            |
| Upper                                              | No data available          |
| Lower                                              | No data available          |
| Vapor Pressure                                     | < 1.0 mmHg @ 25 °C         |
| Vapor Density                                      | 7.25                       |
| Specific Gravity                                   | 1.058                      |
| Solubility                                         | Soluble in water           |
| Partition coefficient; n-octanol/water             | No data available          |
| Autoignition Temperature                           | 420 °C / 788 °F            |
| Decomposition Temperature                          | No information available   |
| Viscosity                                          | 75 mPa.s at 20 °C          |
| Molecular Formula                                  | C8 H19 O4 P                |
| Molecular Weight                                   | 210.21                     |
| Self-Accelerating Decomposition Temperature (SADT) | >75°C                      |

## 10. Stability and reactivity

|                                         |                                                                                          |
|-----------------------------------------|------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                               |
| <b>Stability</b>                        | Stable under recommended storage conditions.                                             |
| <b>Conditions to Avoid</b>              | Keep away from open flames, hot surfaces and sources of ignition. Incompatible products. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents                                                                  |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Oxides of phosphorus            |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                 |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                            |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component         | LD50 Oral         | LD50 Dermal | LC50 Inhalation |
|-------------------|-------------------|-------------|-----------------|
| Dibutyl phosphate | >2000 mg/kg (Rat) | Not listed  | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | Causes burns by all exposure routes                                                      |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component         | CAS No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------|----------|------------|------------|------------|------------|------------|
| Dibutyl phosphate | 107-66-4 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated. See actual entry in RTECS for complete information.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

| Component         | Freshwater Algae | Freshwater Fish                            | Microtox   | Water Flea |
|-------------------|------------------|--------------------------------------------|------------|------------|
| Dibutyl phosphate | Not listed       | LC50: > 100 mg/L, 96h static (Danio rerio) | Not listed | Not listed |

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

| Component         | log Pow |
|-------------------|---------|
| Dibutyl phosphate | -0.9    |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN3265  
 Proper Shipping Name CORROSIVE LIQUID, ACIDIC, ORGANIC, N.O.S.  
 Technical Name Dibutyl phosphate  
 Hazard Class 8  
 Packing Group III

#### TDG

UN-No UN3265  
 Proper Shipping Name Corrosive liquid, acidic, organic, n.o.s.  
 Hazard Class 8  
 Packing Group III

#### IATA

UN-No UN3265  
 Proper Shipping Name Corrosive liquid, acidic, organic, n.o.s.  
 Hazard Class 8  
 Packing Group II

#### IMDG/IMO

UN-No UN3265  
 Proper Shipping Name Corrosive liquid, acidic, organic, n.o.s.  
 Hazard Class 8  
 Packing Group II

### 15. Regulatory information

#### United States of America Inventory

| Component         | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|-------------------|----------|------|-----------------------------------------------|-----------------------------|
| Dibutyl phosphate | 107-66-4 | X    | ACTIVE                                        | -                           |

#### **Legend:**

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

**International Inventories**

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component         | CAS No   | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|-------------------|----------|-----|------|-----------|-------|------|------|------|-------|----------|
| Dibutyl phosphate | 107-66-4 | X   | -    | 203-509-8 | X     | X    | X    | X    | X     | KE-28562 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**U.S. Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)** Not applicable

**Clean Air Act** Not applicable

**OSHA - Occupational Safety and Health Administration** Not applicable

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component         | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|-------------------|---------------|------------|--------------|----------|--------------|
| Dibutyl phosphate | X             | X          | X            | -        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

**Other International Regulations**

**Mexico - Grade** No information available

**Authorisation/Restrictions according to EU REACH** Not applicable

| Component | CAS No | REACH (1907/2006) - | REACH (1907/2006) - | REACH Regulation (EC |
|-----------|--------|---------------------|---------------------|----------------------|
|-----------|--------|---------------------|---------------------|----------------------|

|                   |          | Annex XIV - Substances<br>Subject to Authorization | Annex XVII - Restrictions<br>on Certain Dangerous<br>Substances | 1907/2006) article 59 -<br>Candidate List of<br>Substances of Very High<br>Concern (SVHC) |
|-------------------|----------|----------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Dibutyl phosphate | 107-66-4 | -                                                  | -                                                               | -                                                                                         |

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component         | CAS No   | OECD HPV       | Persistent Organic<br>Pollutant | Ozone Depletion<br>Potential | Restriction of<br>Hazardous<br>Substances (RoHS) |
|-------------------|----------|----------------|---------------------------------|------------------------------|--------------------------------------------------|
| Dibutyl phosphate | 107-66-4 | Not applicable | Not applicable                  | Not applicable               | Not applicable                                   |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

**Other International Regulations**

| Component         | CAS No   | Seveso III Directive<br>(2012/18/EC) -<br>Qualifying Quantities<br>for Major Accident<br>Notification | Seveso III Directive<br>(2012/18/EC) -<br>Qualifying Quantities<br>for Safety Report<br>Requirements | Rotterdam<br>Convention (PIC) | Basel Convention<br>(Hazardous Waste) |
|-------------------|----------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------|
| Dibutyl phosphate | 107-66-4 | Not applicable                                                                                        | Not applicable                                                                                       | Not applicable                | Not applicable                        |

**16. Other information****Prepared By**

Regulatory Affairs  
Thermo Fisher Scientific  
Email: EMSDS.RA@thermofisher.com

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**Revision Summary**

This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**